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Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738212897479-Chapter 4 - Unit 2 - Bringing Ideas to Life.pdf

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

HOW TECHNOLOGY HELPS IN FLOOD PREVENTION

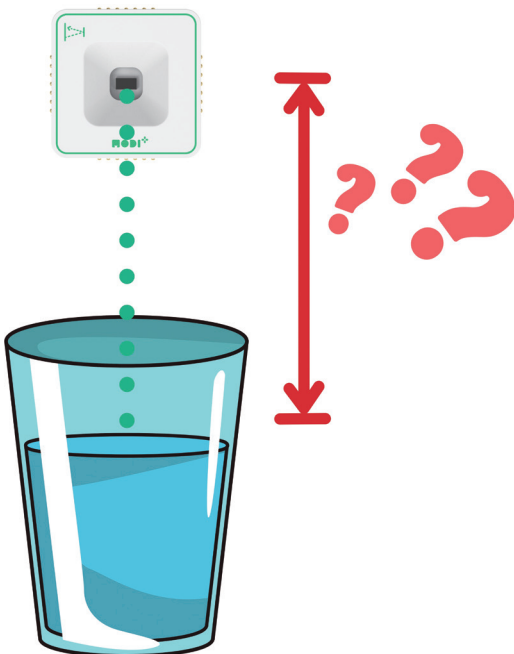
Technology plays a crucial role in preventing and managing floods. By using advanced tools and systems, we can detect rising water levels early, provide warnings to communities, and implement measures to minimize damage.

HOW TECHNOLOGY HELPS IN FLOOD PREVENTION

The Flood Alarm System we are going to build will use a Time-of-Flight (ToF) sensor, a speaker, and LED modules to detect rising water levels and provide alerts. Here's how it works:

1. Detecting Water Levels with ToF Sensor

- The ToF sensor measures the distance between itself and a flat object, such as a piece of paper, placed on the water surface. Since the sensor's light is not reflected by the water, the paper acts as a marker that the sensor can detect.
- As the water level rises, the paper moves closer to the ToF sensor, allowing it to measure the decreasing distance.



In case the ToF sensor's measurement does not work perfectly due to the nature of water, it's better to use a piece of paper as a clear surface.



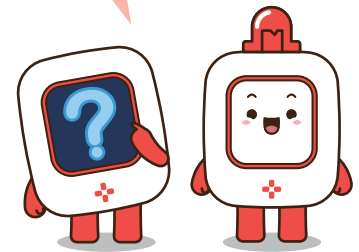


2. Processing Information

- The collected data from the ToF sensor is processed to determine the water level. When the distance between the sensor and the paper reaches a certain threshold, it indicates that the water level is dangerously high.
- Think about how the system could use different ways to alert people once it detects a high water level. Consider both visual and auditory signals. In this project, we will use two visual indicators by using two different modules for it.

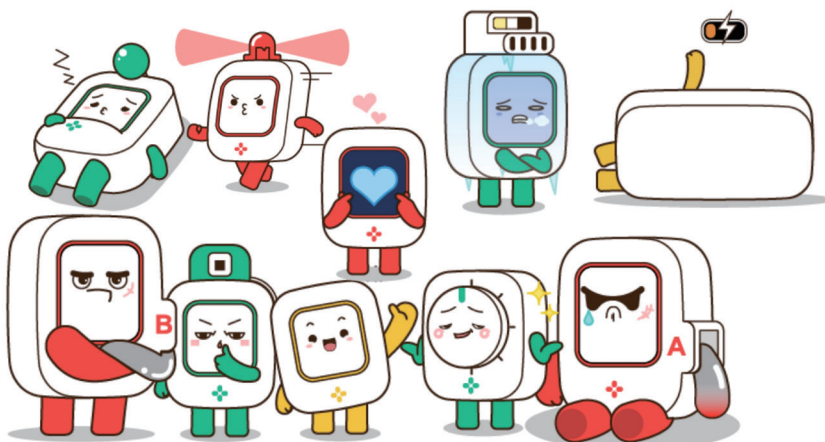
HINT

Among the MODI modules, which ones can be used for visual signals?



3. Providing Feedback

- The system needs to give immediate feedback when a high water level is detected.
- Hint: One module could light up to indicate a warning, another could emit a sound, and another could display a message or status update.

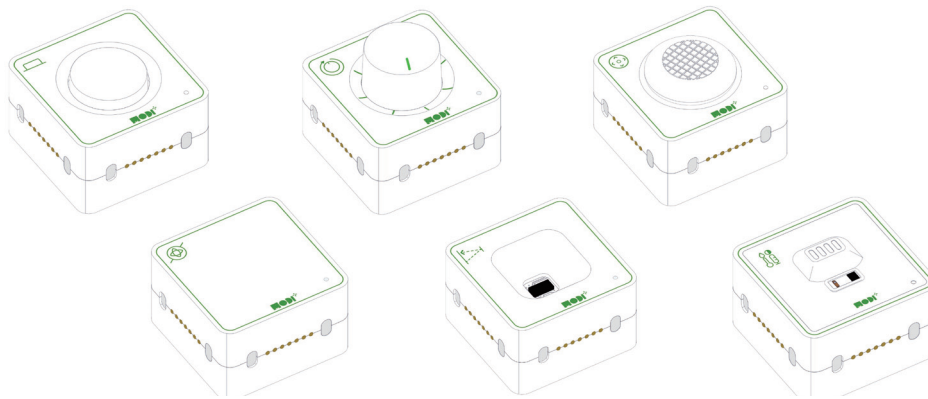


Required Modules & Brainstorming

To build our Flood Alarm System, we need to decide which MODI modules to use. Here are some ideas to get started:

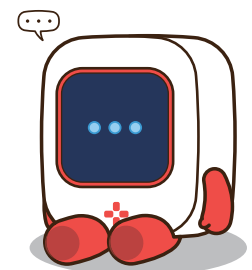
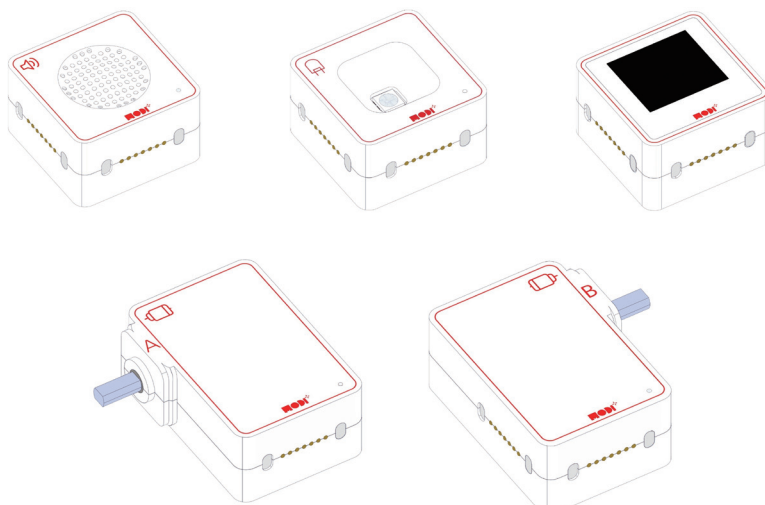
Sensor Modules

- ToF Sensor: Measures the distance between the sensor and a flat object on the water surface to detect rising water levels.
- Flat Object (e.g., Paper): Placed on the water surface to reflect the ToF sensor's light and provide accurate distance measurements.



Output Modules

- Visual Signal Module: Think about which MODI modules can provide visual signals.
- Auditory Signal Module: Consider which module can emit sound to provide an auditory alert.
- Display Module: A module that can show messages or status updates indicating the water level.



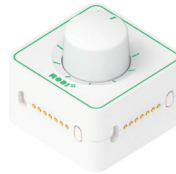


Required Modules & Brainstorming

Which MODI modules do you need for the creation?



BUTTON



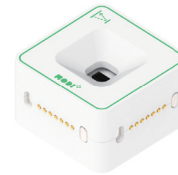
DIAL



ENVIRONMENT



JOYSTICK



TOF



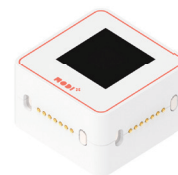
SPEAKER



IMU



LED



DISPLAY

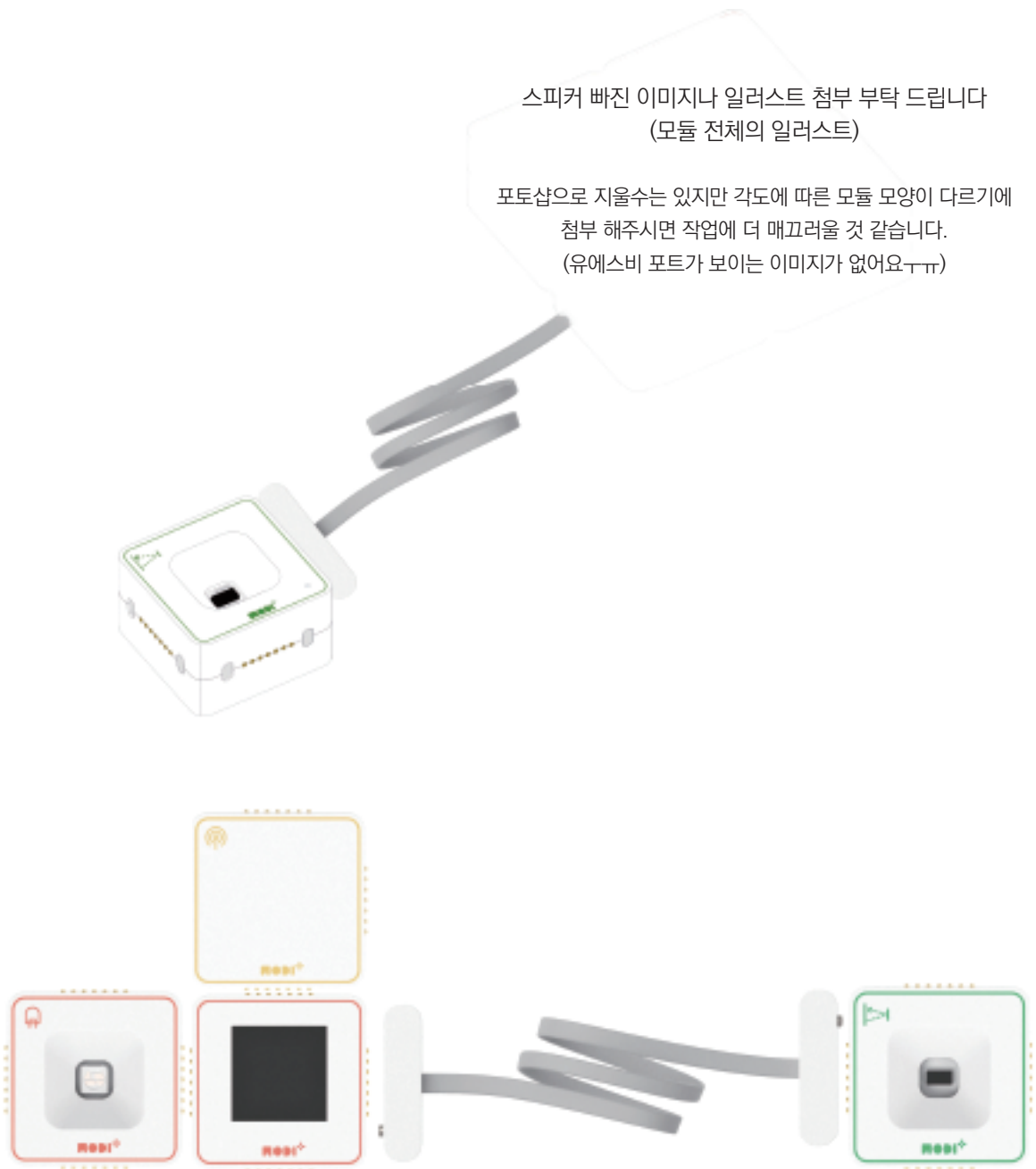


MOTOR A



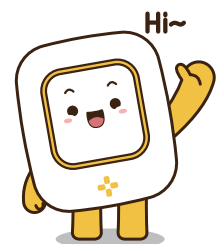
MOTOR B

WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.



TIPS

By using the extension cable, you can connect modules even if they are far apart from each other.





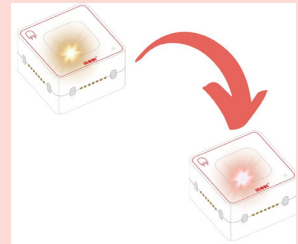
Module Functionality

LED Module

How It Works The LED module is used to provide visual signals. It lights up in different colors to indicate various levels of alerts.

APPLICATIONS IN FLOOD ALARM SYSTEM

- **Warning Indicator** The LED lights up in yellow when the water level is approaching the warning threshold.
- **Danger Indicator** The LED lights up in red when the water level reaches a dangerous level, signaling an immediate alert.



LED Module

How It Works The Display module shows text or graphical information, providing detailed alerts and instructions.

APPLICATIONS IN FLOOD ALARM SYSTEM

- **Status Update** The display shows real-time water level measurements.
- **Alert Messages** When the water level reaches critical points, the display shows messages like "Warning: Water Level Rising" or "Danger: Flood Imminent."

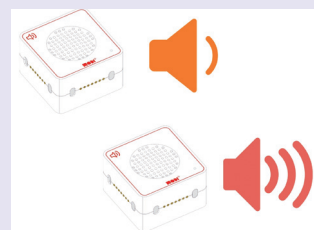


Speaker Module

How It Works The Speaker module emits sound alerts to grab attention and provide auditory warnings.

APPLICATIONS IN FLOOD ALARM SYSTEM

- **Warning Sound** The speaker emits a low sound when the water level is near the warning threshold.
- **Alarm Sound** The speaker emits a loud alarm when the water level reaches a dangerous level, signaling an urgent need to take action.

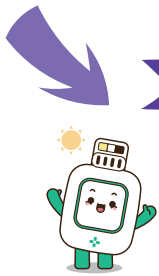


WHY THESE MODULES ARE EFFECTIVE IN A FLOOD ALARM SYSTEM

The ToF sensor is used in many cool ways because it can measure distance very well. Here are some places where you might find it:

Comprehensive Alerts

By combining visual (LED), detailed (Display), and auditory (Speaker) alerts, the warning system is effective in different situations and for various people. Some might see the light first, while others might hear the sound or read the display.

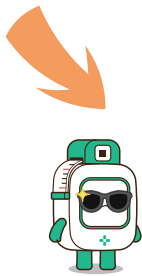


DISCUSSION POINT

Why is it important to have different ways to alert people in alarm systems?
Can you think of one example and share it with your classmates?

Redundancy

Having multiple forms of alerts ensures that if one method is missed, others will still convey the necessary warnings.

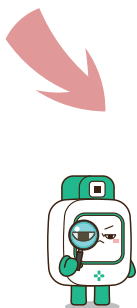


DISCUSSION POINT

Why is it important to have different ways to alert people in alarm systems?
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Accessibility

The system is more accessible to people with different sensory abilities. For instance, visual alerts help those who are hearing impaired, while sound alerts assist those who might not be looking at the display.



DISCUSSION POINT

Our society includes many different people, some with impaired sight or hearing. Think of one example where using different types of alerts can help people with sensory disabilities.